Storage Unit

Storage Unit



















Bus

Storage Unit

- The storage unit is a part of the computer system which is employed to store the **data and instructions to be processed.**

Bus

Types of Storage Devices

- Primary Storage Devices
- Secondary Storage Devices

Bus

Storage of data

1 Kilobyte (KB) =1024 bytes

1MegaByte (MB)= 1024 KB

1GigaByte (GB) =1024 MB

1TeraByte(TB)=1024 GB

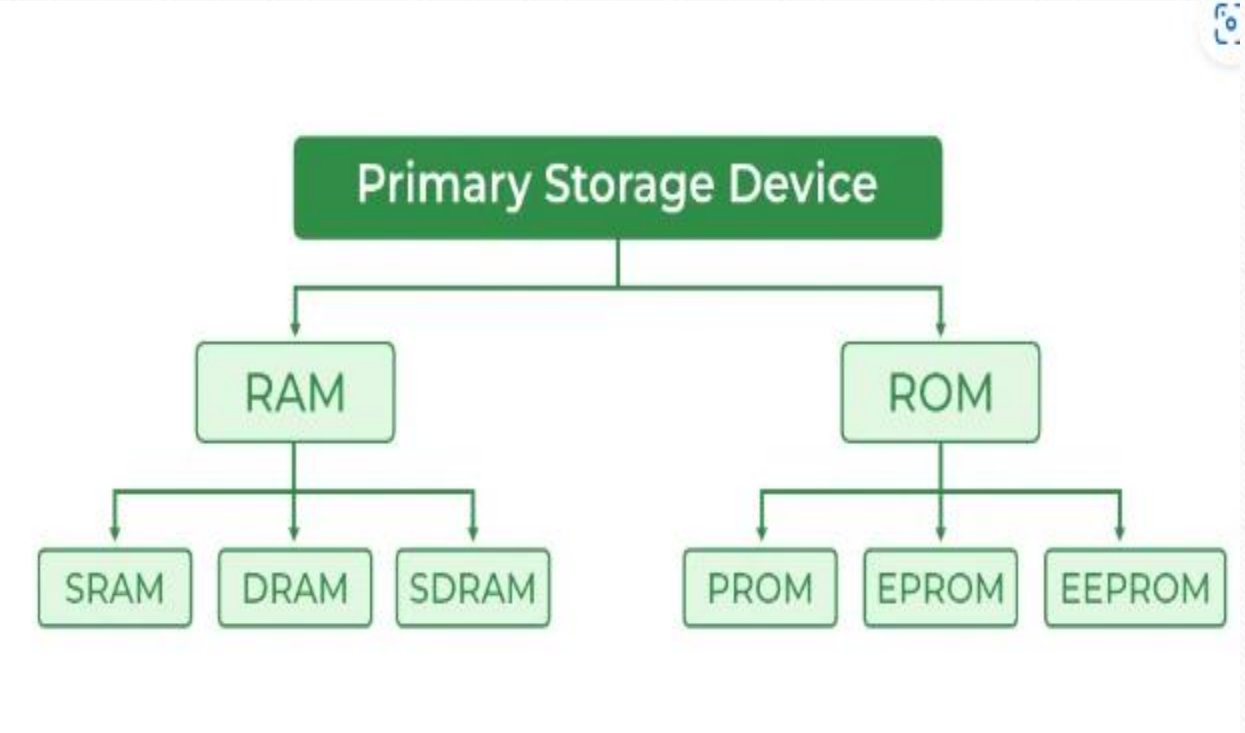
1 PetaByte (PB) =1024 TB

Bus

Primary Memory

- It is also known as **internal memory** and **main memory.**
- This is a section of the CPU that holds program **instructions, input data, and intermediate results.**
- It is generally **smaller in size.**

Primary Storage Devices



Bus

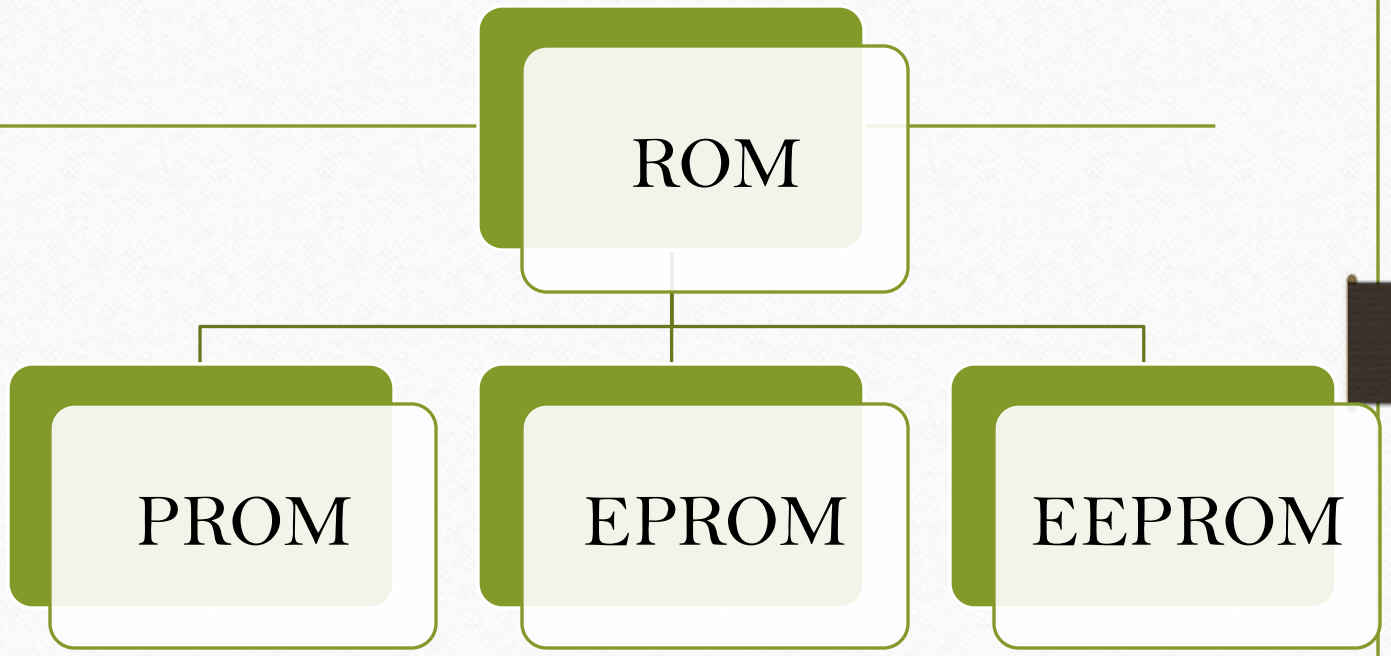
- It stands for **Read-Only Memory**.
- ROM is a **non-volatile memory**.

ROM

- The data stored in the memory cannot be modified or deleted.
- The data is stored permanently during manufacturing time.
- ROM stores instructions that are used to start a computer.
- This operation is referred to as **bootstrap.**

Bus

Types of ROM



Bus

PROM

- PROM is **Programmable Read-Only Memory.**
- These are ROMs that can be programmed.
- A special PROM programmer is employed to enter the program on the PROM.
- Once the chip has been programmed, information on the PROM can't be altered.

Bus

EPROM

- EPROM stands for **Erasable Programmable Read-Only Memory.**
- The data can be erased by passing ultra violet rays into it.
- While passing the rays the entire data will be removed

Bus

EEPROM

- EEPROM is **Electrically erasable programmable read-only memory.**
- Here, data can be erased by applying electrical signals into it.
- We can erase data bit by bit in it

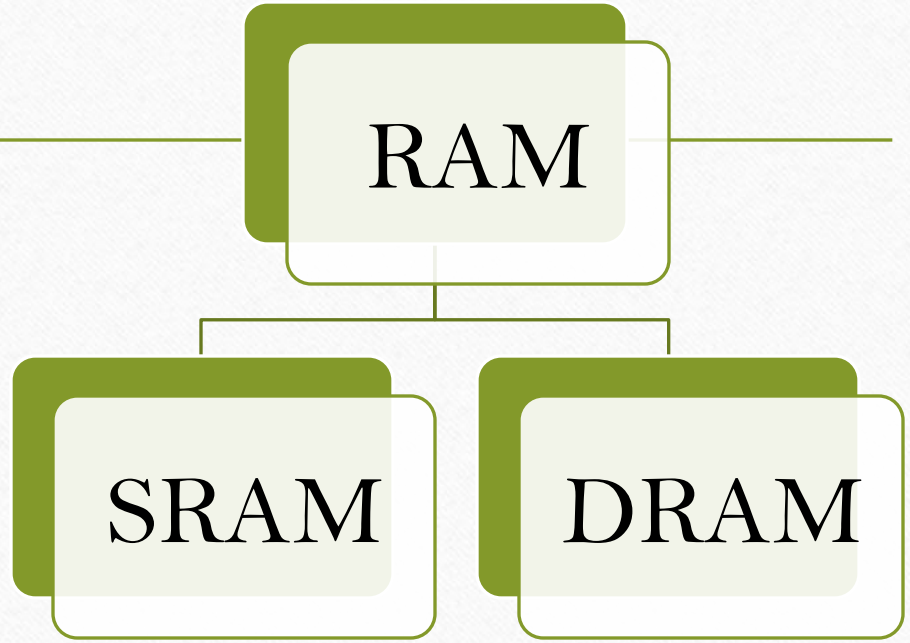
Bus

RAM

- It stands for **Random Access Memory**.
- It is a **volatile memory**
- Ie, the data will be lost when the power is off
- All the currently running applications are stored in RAM

Bus

Types of RAM



Bus

SRAM

- It stands for **Static Random Access Memory.**
- It consists of circuits that retain stored information as long as the power supply is on.
- The transistors are used to store the data

Bus

DRAM

- It stands for **Dynamic Random Access Memory.**
- It is used to store binary bits in the form of electrical charges that are applied to capacitors.
- The access time of DRAM is slower as compared to SRAM but it is cheaper than SRAM and has a high packaging density.

Bus

SRAM	DRAM
Transistors are used to store information in SRAM.	Capacitors are used to store data in DRAM.
Capacitors are not used hence no refreshing is required.	To store information for a longer time, contents of the capacitor needs to be refreshed periodically.
SRAM is faster as compared to DRAM.	DRAM provides slow access speeds.
These are expensive.	These are cheaper.
SRAMs are low density devices.	DRAMs are high density devices.
These are used in cache memories.	These are used in main memories.

Secondary Memory

- Secondary memory is a type of computer memory that is used to store data and programs permanently
- It is non-volatile and can store data and programs for extended periods of time.

Bus

Types of Secondary Memory

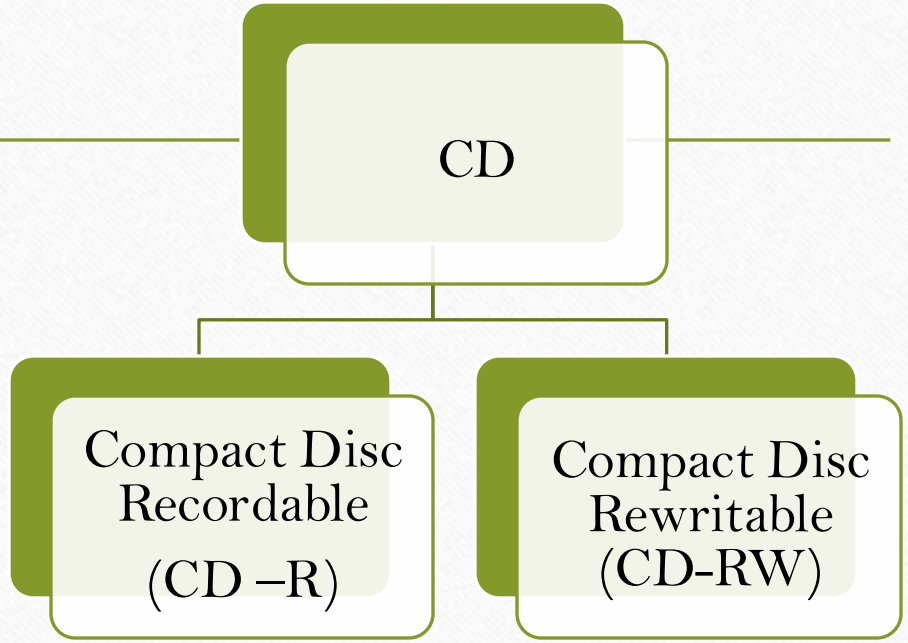
- The commonly used secondary storage devices includes
  - CD
  - DVD
  - Hard disk
  - Blue ray disk
  - Flash Drive
  - Memory Stick
  - Smart Card

Compact Disc

- A **Compact Disc** (CD) is a commonly used secondary storage device.
- It contains **tracks and sectors** on its surface.
- Its shape is circular and is made up of polycarbonate plastic.
- The storage capacity of CD is up to 700 MB of data.

Bus

Types of CD



CD-R : Once the data has been written onto it cannot be erased, it can only be read.

CD-RW : It is a special type of CD in which data can be erased and rewritten as many times as we want.

DVD

- Stands for **Digital Versatile Disc**
- It looks just like a CD, but the storage capacity is greater compared to CD, it stores up to 4.7 GB of data.
- DVD-ROM drive is needed to use DVD on a computer.
- The video files, like movies or video recordings, etc., are generally stored on DVD and we can run DVD using the DVD player

Blue Ray Disk

- Blu-ray is an optical disc data-storage format that is most often used for playback of high-definition (HD) video
- A Blu-ray disc looks just like a CD or a DVD but it can store data or information up to 25 GB data.
- If we want to use a Blu-ray disc, we need a Blu-ray reader.

Flash Drive

- A flash drive or pen drive comes in various storage capacities, such as 1 GB, 2 GB, 4 GB, 8 GB, 16 GB, 32 GB, 64 GB, up to 1 TB.
- A flash drive is used to transfer and store data.
- To use a flash drive, we need to plug it into a USB port on a computer.

Memory Stick

- A memory stick is a flash memory card manufactured for use with handheld devices.
- Memory sticks are small, portable devices that allow you to transport your files from one computer to another easily.

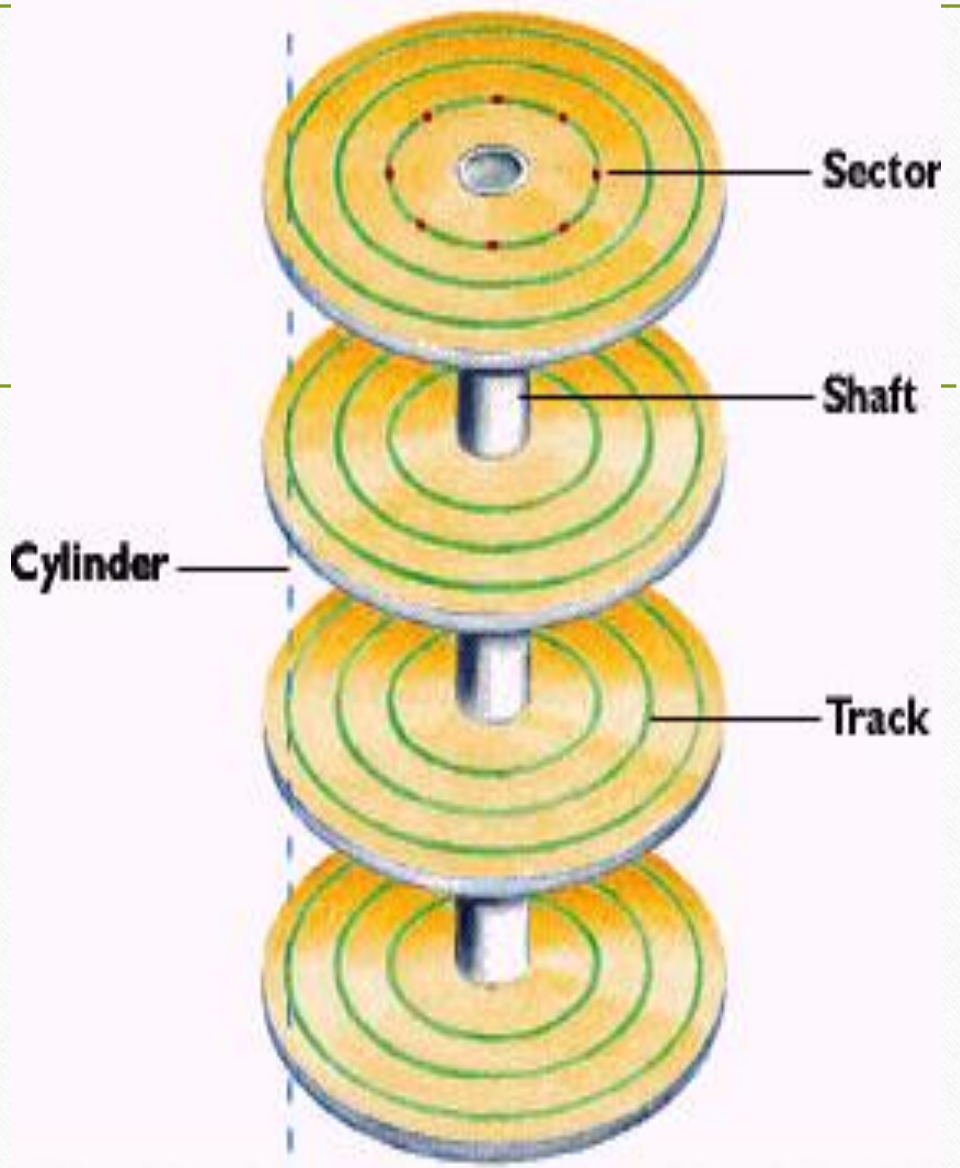
Smart Card

- It is typically a plastic credit card-sized card with an embedded integrated circuit (IC) chip
- Smart cards can provide personal identification, authentication, data storage, and application processing.

Hard Disk

- A hard disk is often called a **disk drive, hard drive or hard disk drive**
- Stores and provides relatively quick access to large amounts of data on an electromagnetically charged surface or set of surfaces.

Physical Structure of Hard Disk



Physical Structure of Hard Disk

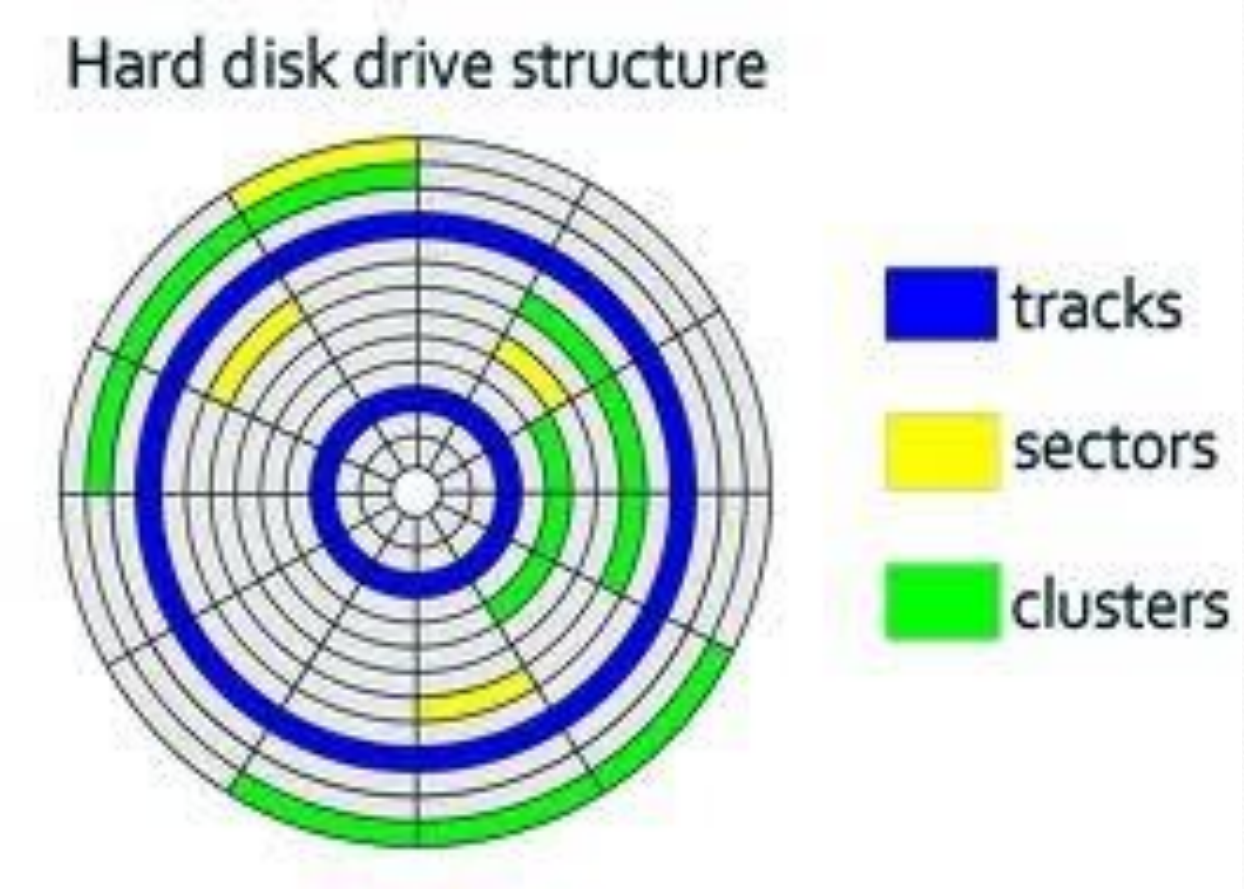
- Inside of a hard disk there is the spindle which is used to hold **circular magnetic disks** also known as **platters**.
- They are usually made up of a non magnetic material, like aluminum alloy.

Physical Structure of Hard Disk

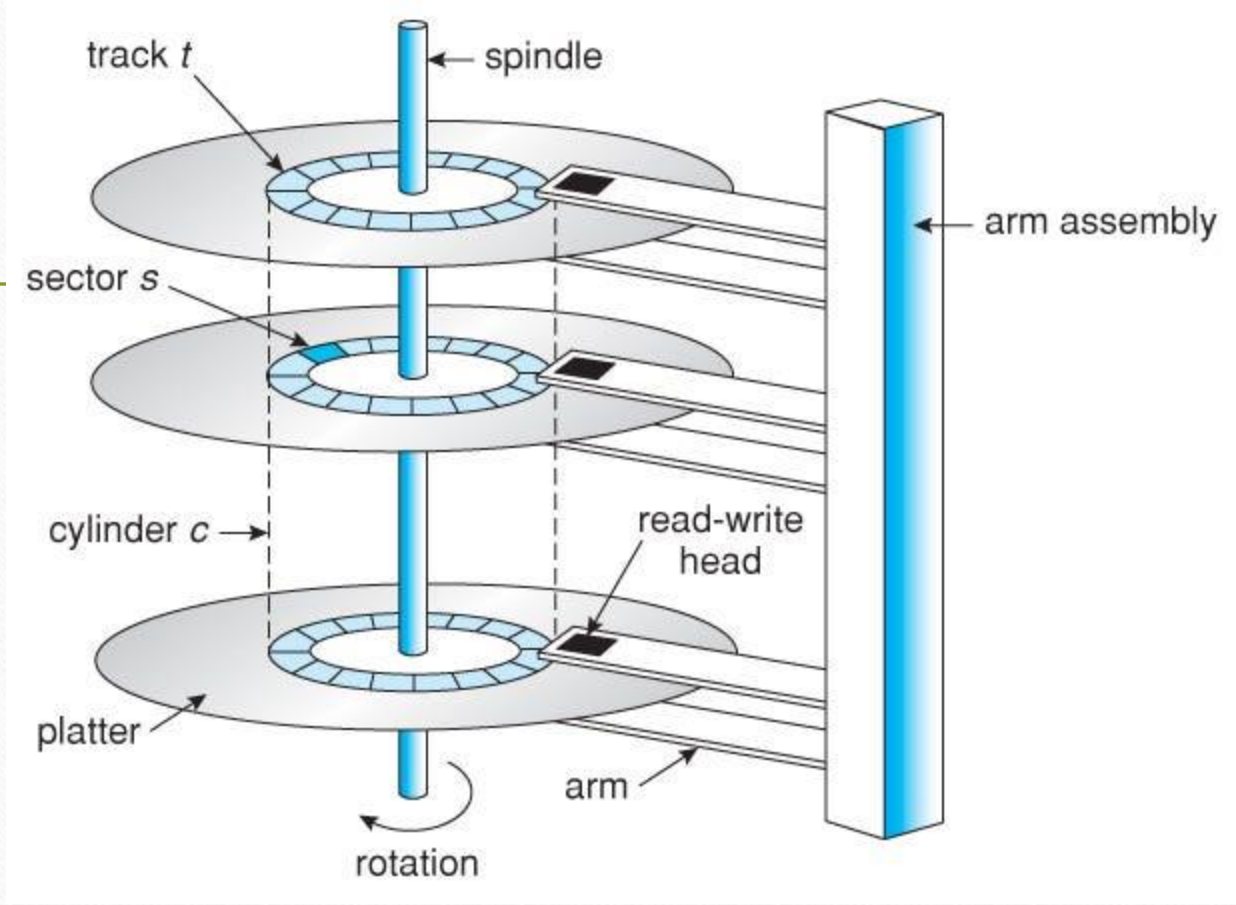
Physical Structure of Hard Disk



Physical Structure of Hard Disk



Physical Structure of Hard Disk



Physical Structure of Hard Disk

- Each platter is divided into thin concentric bands known as **tracks**.
- There can be more than a thousand tracks on a 3.5 inch hard disk.
- The tracks are further subdivided into **sectors**.
- These are the smallest physical storage unit on a disk and they are almost always 512 bytes long.
- Group of sectors are called **Cluster**.

Physical Structure of Hard Disk

- The platters are sealed inside a container
- The container contains a motor to rotate the disk.
- A hard disk drive typically consists of two read-write heads for each platter disk. One of the heads is attached to the top of the platter, whereas, the other is placed in close contact

Reading and writing from hard disk

- The R/W head reads and writes data to a hard disk.

Features of hard disk

- **Seek time**: The time required to move the read write heads to the specified track
- **Latency**: the time is taken by the sector to come under the R-W head.
- **Access Time** : the time taken to find the required data